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The effect of stocking density, flock size and modified management on laying hen behaviour and welfare in a non-cage system

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Abstract

The current large-scale experiment aimed to study laying hen behaviour under commercial stocking densities, flock sizes and management practices using a replicated design.

Thirty-six flocks of beak-trimmed Shaver laying hens (113,400 birds in total), six flocks per treatment, were housed within commercial single-tier aviary systems. The six treatments comprised different combinations of stocking densities (low: 7 birds m⁻², medium: 9 birds m⁻², high: 12 birds m⁻²), flock sizes (small: 2450/3150 birds, large: 4200 birds) and management conditions (standard and modified). Bird behaviour (incidence of feather pecking, aggression, preening, dustbathing and allopreening) was recorded directly by an observer when birds were approximately 32, 48 and 60 weeks of age.

The initial level of feather pecking and aggression was highest in the low stocking density. Feather pecking and aggression increased with age but only in the high stocking density treatments. In the high stocking density treatments more aggression, preening and allopreening were recorded in small flocks than in large flocks and especially the small flocks under standard management conditions showed higher levels of feather pecking and aggression by the end of the laying cycle. This effect of

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small, high density flocks on feather pecking and aggression was counteracted by modified management conditions.

Behavioural observations in this study did not show that the welfare of laying hens was compromised by housing them at 12 birds m^{-2} , in comparison with birds housed at 9 or 7 birds m^{-2} in single-tier aviary system. However, modifications in management decreased feather pecking and aggression.

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1. Introduction

As the 2012 EU ban on conventional cages for laying hens approaches, non-cage or alternative systems are likely to become more predominant. This has raised the issue of what stocking density is most appropriate for production and welfare of hens. The Council Directive 1999/74/EC (C.o.t.E., 1999), which lays down minimum standards for the welfare of laying hens, states a maximum stocking density of 9 birds m^{-2} . However, where the usable area corresponds to the available ground surface a stocking density of 12 hens m^{-2} is allowed until 2012 in systems applying this density on 3 August 1999. There is no clear scientific evidence for the recommendation of the maximum stocking density of 9 birds m^{-2} , because large-scale replicated experimental studies are few. Whether or not the welfare of hens is improved by housing birds at 9 animals m^{-2} , or less, can only be assessed by considering the effect of stocking density on a broad range of different potential indicators of welfare. One of these indicators is laying hen behaviour.

Higher stocking densities have been associated with higher levels of feather pecking in laying hens, in different systems (Nicol et al., 1999). Also laying growers (Hansen and Braastad, 1994; Huber-Eicher and Audige, 1999) and growing bantams (Savory et al., 1999) showed more feather pecking, measured by the amount of feather damage, under higher densities. Furthermore, epidemiological studies have shown that use of the outdoor range in free-range flocks of laying hens reduced the risk of feather pecking (Bestman and Wagenaar, 2003; Nicol et al., 2003), which might be the result of lower stocking densities within the barn. Other studies, however, failed to find a relationship between stocking density and feather pecking (Appleby et al., 1989; Carmichael et al., 1999; Gunnarsson et al., 1999; Oden et al., 2002).

Under commercial conditions, an increase in stocking density is often accompanied by an increase in flock size. An increase in flock size is associated with higher levels of feather pecking (Nicol et al., 1999; Bilčík and Keeling, 2000), higher levels of fear (Bilčík et al., 1998) and lower bodyweight (Keeling et al., 2003). However, levels of aggression are generally low in large flocks of laying hens (Hughes et al., 1997; Carmichael et al., 1999; Nicol et al., 1999; Estevez et al., 2002, 2003), possibly due to the formation of sub-groups of familiar individuals within a large flock (Grigor et al., 1995). Alternatively, low levels of aggression could be the result from a lack of social relationships brought about by a strategy of social tolerance (Hughes et al., 1997; Pagel and Dawkins, 1997; Estevez et al., 2003).

In the studies mentioned above a wide range of stocking densities (1.3–30 birds m⁻²) and flock sizes (10–5000 birds) were used which makes it difficult to draw unequivocal conclusions about the impact of stocking density and flock size on laying hen behaviour and welfare. This is also complicated by the fact that very few of these studies have used a replicated design. Furthermore, the flock sizes used were usually at the low end of the scale relative to commercial conditions which makes it difficult to extrapolate findings to a commercial scale. Also, different studies used different housing systems and types of management, and stocking density and flock effects were confounded across studies. Management factors, such as, for example, application of lights in nest boxes, litter management, quality of outdoor range and type of drinker used, have a substantial impact on chicken behaviour (Green et al., 2000; Pöttsch et al., 2001; Nicol et al., 2003).

The aim of the current study was to examine the differential effects of stocking density and flock size on the behaviour of laying hens under commercial conditions. The effects of management interventions, designed to reduce the risk of feather pecking, were additionally evaluated for hens kept at the highest stocking density. The effects of these factors on bird mortality, physical condition and physiology are reported separately (Nicol et al., *in press*). The benefits of working on a commercial farm were tempered by practical constraints, which limited the independence of some replicates, and produced house- $\times$ treatment confounding. Despite these difficulties, we aimed to study laying hen behaviour under current commercial stocking densities, flock sizes and management practice with a greater degree of replication than is normally achieved in such trials.

2. Methods

In order to study the effects of stocking density, flock size and management on behaviour of laying hens (Shavers) in non-cage systems, the following six treatments were assigned to six structurally identical barns located at a commercial farm in the UK (Table 1).

Under standard management (SM) conditions pens were equipped with bell drinkers and lights were on in the nest boxes. In the treatments with a modified management (MM), the birds had nipple lines instead of bell drinkers and there were no lights in the nest boxes. These modifications have been associated with decreased risks of feather and vent pecking in epidemiological studies (Green et al., 2000; Pöttsch et al., 2001).

Table 1
Stocking density, flock size and type of management in the six treatments

Treatment	Stocking density (birds m ⁻²)	Flock size (no. of birds)	Management type
1	7	Small (2450)	Standard (SM)
2	9	Small (3150)	Standard
3	12	Large (4200)	Standard
4	12	Large (4200)	Modified (MM)
5	12	Small (2450)	Standard
6	12	Small (2450)	Modified

For description of the management types, see text.

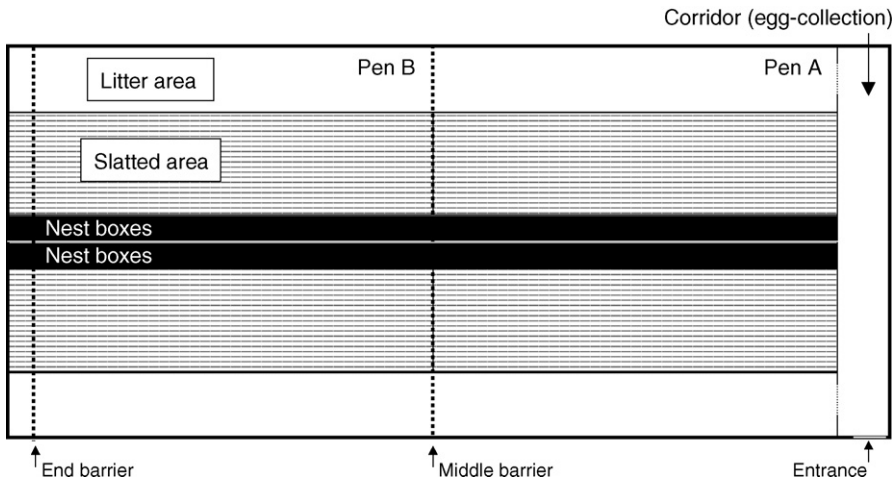


Fig. 1. Schematic drawing of a barn. The slatted platform contained feeders and drinkers. Nest boxes ran along the whole length of the barn.

Each barn was divided in two separate pens (pens A and B) by a wire mesh barrier (see Fig. 1). Each pen contained one replicate of a treatment. So, for example, the barn with treatment 1 contained a total of 4900 birds (2450 in each pen) at the start of the treatment. Three rounds per treatment were run in total, resulting in six replicates per treatment (making up a total of 113,400 birds in this study).

The length of the pens differed between treatments. Each pen (measuring a maximum of 30 m × 14 m for the large flocks) consisted of a litter area (width ±2.20 m) and a raised (±1 m high) wooden slatted platform (width ±3.70 m), along the length of the barn. Two rows of nest boxes subdivided a pen into two sides. Both sides were identical in layout but mirrored. The slatted platform contained feeders and drinkers along the whole length of the pen.

In all treatments individual hens had on average 10 cm feeder space. Under standard management conditions one bell drinker per 88 birds was available. Under modified management conditions one nipple/cup was available for every 10 hens. One nest was provided for every seven hens. The nests were provided in one tier in treatments 1 and 2 and in two tiers in the other treatments.

The target ambient temperature was 20 °C. The light period ran from 05:00 to 19:00 h. All barns were equipped with fluorescent tubes supplemented by tungsten bulbs to provide a light environment free from bright spots and shadows. The target light intensity at the nest box entrance was 10 lx. Under standard management lights in the nest boxes were on between 04:00 and 18:00 h.

At approximately 70 weeks of age the barns were depopulated. Barns were cleaned and prepared for the next round.

Due to logistical difficulties on the farm not all treatments were run simultaneously; there was a maximum gap of about 6 months between treatments. It was also not possible to randomly assign treatments to barns, because of financial and practical constraints, e.g. the expensive additional tiers of nest boxes required for the higher stocking density treatments

could only be installed in some houses. So, treatment was totally confounded with barn, except for treatment 5 in which the first four replicates were run at the same time in two separate barns (this was due to a delay in populating the barn for the first round, caused by a fire in the rearing flock), and treatment 3 in which the last two replicates were run in two separate barns. However, close monitoring of environmental conditions at the start and throughout the experiment was undertaken to ensure that any apparent barn effects were considered.

3. Behavioural observations

Behavioural observations were carried out at three time points for each flock, at approximately 3, 7 and 10 months into the laying cycle, when the birds were about 32, 48 and 60 weeks of age, respectively. For the behavioural observations at each time point a pen was subdivided into 16 virtual areas of similar length, 8 on the litter and 8 on the slats. The boundaries of an area were marked by features in the environment. The areas were scored in succession. The order of areas was determined beforehand. After entering a pen the observer slowly walked to the position from which the first observation would be carried out and waited 5 min to accustom the birds to the human presence. All observations were carried out from the edge of the slats, next to the litter area. Before and after each observation the number of birds in the area under study was counted. These two values were used to calculate the mean number of birds in an area during an observation. In a 5-min observation period, all occurrences of the following behaviours were scored on a check-sheet by slowly scanning all birds present in that area:

- *Feather pecking*: pecking at or pulling out the feathers of another hen. No distinction was made between gentle and severe feather pecking (Savory, 1995). Nearly all pecks observed were gentle, though;
- *Aggression*: vigorous pecks at the neck or head of another bird, threats, chases, attacks and fights;
- *Preening*: this included feather shaking, wing/leg stretching and wing flapping;
- *Dustbathing*;
- *Allopreening*: this was defined as gentle pecks aimed at the beak or face or at particles in the plumage of another hen. The latter is often seen in a dustbathing context.

Behaviours were expressed as incidence per bird per hour. The majority of observations was carried out by two observers (PHZ and ACL). On a few occasions two other observers assisted in carrying out the observations.

4. Data analysis

Data of the 16 areas per pen were averaged. Therefore, pen was the statistical unit. An initial comparison between overall levels of feather pecking and aggression in all six treatments (T1–T6) was made.

Statistical analysis was then performed using the General Linear Methods procedure in SAS (version 6.12). Square root transformations were applied for skewed distributions of frequency.

The effect of stocking density was examined using only the small flocks under standard management conditions (T1, T2, T5). Effects of and interactions between the factors stocking density (7, 9 or 12 m⁻²) and age (32, 48 or 60 weeks) were assessed using variance analysis; the effect of round was included in the model. A split-plot model was used with values in time of individual pens taken as repeated measurements; i.e., effects of stocking density and round were tested against the effect of pen within stocking density and round.

Only the high density treatments (T3–T6) were used to analyse the effects of and interactions between flock size (small or large), management (standard or modified), age (32, 48 or 60 weeks) and their interactions; round was included in the model. A split-plot model was used with values in time of individual pens taken as repeated measurements; i.e., effects of flock size, management, their interaction and round were tested against the effect of pen within flock size, management and round.

Significant effects were further analysed using post hoc tests with Bonferroni adjustments for multiple comparisons. For brevity, only main effects and significant interaction effects are reported in the text. Data are presented as means $\pm$ S.E.M.

5. Results

Fig. 2 shows the overall effect of treatment on the incidence of feather pecking ($F_{5,59} = 2.9$, $P < 0.05$, top figure) and aggression ($F_{5,59} = 5.3$, $P < 0.01$, bottom figure).

On average, feather pecking and aggression were highest in the small flocks under standard management and lowest in treatments under modified management.

5.1. The effect of stocking density in small flocks under standard management conditions (T1, T2, T5)

We found a significant effect of the interaction stocking density $\times$ age ($F_{2,29} = 3.4$, $P < 0.05$; Fig. 3) on the incidence of feather pecking. In the low (7 birds m⁻²) and medium (9 birds m⁻²) stocking density pens the incidence of feather pecking was not affected by age. In high stocking density pens more feather pecking was shown at 48 and 60 weeks than at 32 weeks of age ($P < 0.01$ and < 0.001 , respectively), and more feather pecking was performed at 60 than at 48 weeks of age, although not significantly so ($P = 0.09$). Analysis per age (32, 48 or 60 weeks) revealed that at 32 weeks of age more feather pecking ($P < 0.01$) was recorded in the low density compared to the high density pens ($F_{2,13} = 5.5$, $P < 0.05$; Fig. 3). Incidences of feather pecking at 48 and 60 weeks of age were not significantly affected by stocking density.

The interaction stocking density $\times$ age significantly affected aggression ($F_{2,29} = 2.9$, $P < 0.05$); see Fig. 4). In the low stocking density pens, more aggression was shown at 48 weeks of age compared to 32 weeks of age ($P < 0.05$) and the level of aggression did not differ between weeks 48 and 60 ($P = 0.28$). In medium stocking density pens, the incidence of aggression tended to be higher at 60 weeks of age than at 32 weeks of age ($P = 0.07$) and

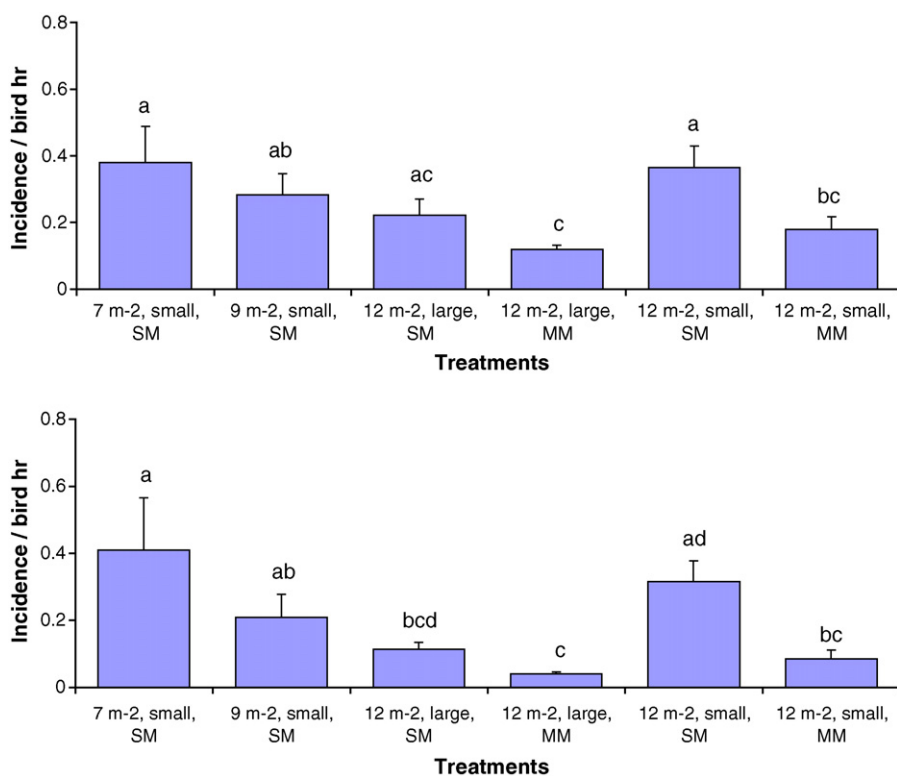


Fig. 2. The incidence of feather pecking (top graph) and aggression (bottom graph) in all treatments. Different letters indicate significant difference below the 0.05-level. ‘Small’ and ‘large’ on the x-axis refers to flock size, SM is standard management, MM is modified management.

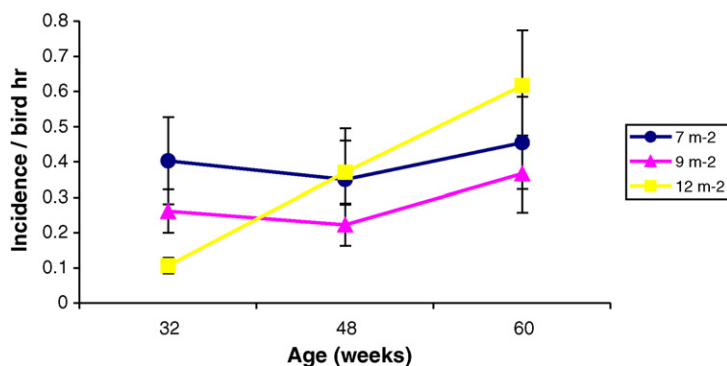


Fig. 3. Incidence of feather pecking at the three stocking densities in the small flocks under standard management conditions. See text for statistics.

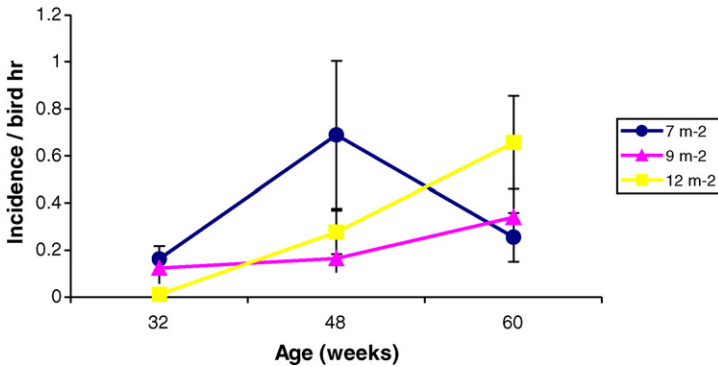


Fig. 4. Incidence of aggression at the three stocking densities in the small flocks under standard management conditions. See text for statistics.

did not differ between weeks 48 and 60 ($P = 0.20$). At the highest stocking density, aggression increased with age and was lower at 32 weeks than at 48 ($P < 0.01$) and 60 weeks of age ($P < 0.001$) and tended to be lower at 48 weeks than at 60 weeks of age ($P = 0.05$).

Analysis per age revealed that at 32 weeks of age more aggression ($F_{2,13} = 5.2$, $P < 0.05$) was recorded at the low and medium stocking densities than at the high stocking density (see Fig. 4).

The incidence of dustbathing was not significantly affected by stocking density (7 m⁻²: 0.14 ± 0.07 , 9 m⁻²: 0.23 ± 0.16 , 12 m⁻²: 0.20 ± 0.09 ; $P = 0.27$), age (32 weeks: 0.14 ± 0.06 , 48 weeks: 0.23 ± 0.15 , 60 weeks: 0.20 ± 0.09 ; $P = 0.41$) or their interaction ($P = 0.72$).

The incidence of preening was affected by stocking density ($F_{2,29} = 10.0$, $P < 0.01$). More preening was shown in the low ($P < 0.05$) and high ($P < 0.001$) density pens compared to the medium stocking density pens. The incidence of preening tended to be higher ($P = 0.06$) at the high density than at the low stocking density (see Fig. 5). Neither

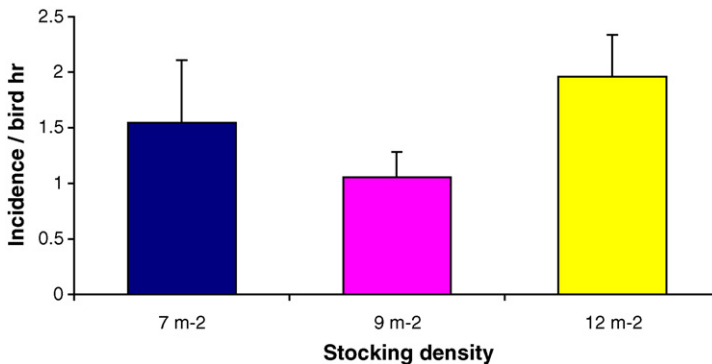


Fig. 5. Incidence of preening at the three stocking densities in the small flocks under standard management. See text for statistics.

age ($P = 0.16$) nor the stocking density $\times$ age interaction ($P = 0.41$) affected the incidence of preening.

The incidence of allopreening was not affected by stocking density (7 birds m^{-2} : 0.34 ± 0.06 ; 9 m^{-2} : 0.28 ± 0.05 ; 12 m^{-2} : 0.43 ± 0.07 ; $P = 0.13$) or age (32 weeks: 0.35 ± 0.08 , 48 weeks: 0.36 ± 0.10 ; 60 weeks: 0.29 ± 0.02).

5.2. The effect of flock size and management in high stocking density flocks (T3–T6)

Feather pecking was affected by the interaction between flock size and age ($F_{2,40} = 5.6$, $P < 0.01$). In the large flocks, the incidence of feather pecking was not affected by age ($P > 0.05$). In the small flocks, however, more feather pecking was shown in week 48 than in week 32 ($P < 0.001$) and more feather pecking was performed in week 60 compared to weeks 32 and 48 ($P < 0.001$ and < 0.05 , respectively; see the solid squares in Fig. 6, top graph).

Feather pecking was also affected by management ($F_{1,18} = 7.9$, $P < 0.05$): more feather pecking was shown under standard management (0.29 ± 0.05) than under modified management conditions (0.15 ± 0.02). The management $\times$ stocking density ($P = 0.64$),

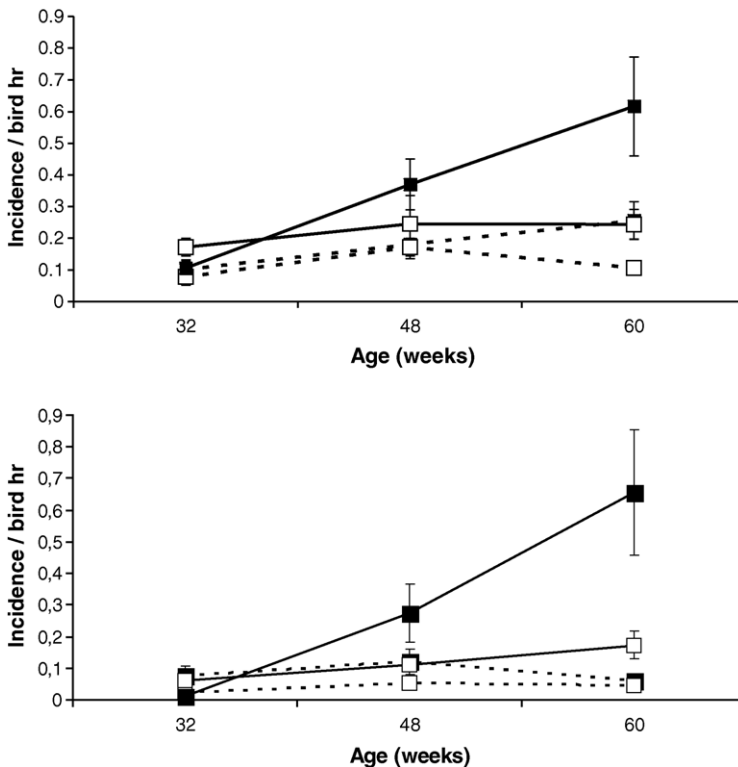


Fig. 6. Incidence of feather pecking (top) and aggression (bottom) in pens with small (solid squares) and large (open squares) flock sizes under the two management conditions (standard: solid line, modified: cross-hatched line). See text for statistics.

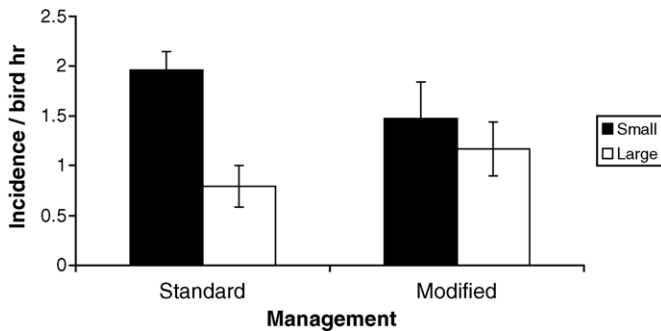


Fig. 7. The mean incidence of preening in the small and large flocks under the two types of management. The solid bars are the small flocks, the open bars represent the large flocks.

management $\times$ age ($P = 0.25$) and management $\times$ stocking density $\times$ age interactions ($P = 0.13$) did not affect the incidence of feather pecking (see top graph in Fig. 6).

The level of aggression was affected by the flock size $\times$ management $\times$ age interaction ($F_{2,40} = 6.6$, $P < 0.01$; see bottom graph, Fig. 6). We found no statistically significant differences in the incidence of aggression between weeks in both the small and large flocks under modified management conditions. In the large flocks under standard management conditions birds tended to show more aggression in week 60 compared to week 32 ($P = 0.06$). In the small flocks under standard management we found a significant increase in aggression over time: birds showed more aggression in week 48 than in week 32 ($P < 0.001$) and more aggression in week 60 than in weeks 32 and 48 ($P < 0.01$ and < 0.001 , respectively).

Dustbathing was not affected by flock size (small: 0.13 ± 0.07 ; large: 0.08 ± 0.03 ; $P = 0.67$), management (standard: 0.10 ± 0.06 ; modified: 0.12 ± 0.04 ; $P = 0.44$), age ($P = 0.24$) or their interactions.

Birds performed more preening ($F_{1,18} = 26.2$, $P < 0.001$) in the small (1.71 ± 0.28) compared to the large flocks (0.98 ± 0.24). We also found a significant effect of the interaction flock size $\times$ management ($F_{2,40} = 10.58$, $P < 0.01$) on the incidence of preening. More preening ($P < 0.001$) was shown in the small than in the large flocks, but only under standard management conditions (Fig. 7). Preening was not significantly affected by the main effect of management (standard: 1.38 ± 0.19 ; modified: 1.32 ± 0.32 ; $P = 0.79$) or age ($P = 0.39$).

More allopreening ($F_{1,18} = 27.90$, $P < 0.001$) was recorded in the small flocks (0.36 ± 0.05) than in the large flocks (0.15 ± 0.01). The incidence of allopreening tended ($F_{1,18} = 3.40$, $P = 0.08$) to be higher under standard management (0.30 ± 0.04) than under modified management conditions (0.21 ± 0.02).

6. Discussion

The aim of the current study was to examine the differential effects of stocking density, flock size and modified management on the behaviour of laying hens under commercial,

large-scale conditions. The effects on physiological and production parameters are reported elsewhere (Nicol et al., *in press*). In the current study, both feather pecking and aggression started off at a higher level in the low (7 birds m⁻²) and medium (9 birds m⁻²) stocking density treatments compared to the high (12 birds m⁻²) density treatment. However, only in the high stocking density treatment feather pecking and aggression showed an increase with age.

Because of practical constraints of working on a commercial farm, treatment and house were partly confounded. Therefore, we closely monitored environmental conditions, such as ammonia concentrations, light intensity, relative humidity and barn temperature, throughout the experiment to ensure that any apparent barn effects were considered. Formal statistical analysis could not be conducted on environmental conditions, which were independent at the barn, not flock, level, providing insufficient replication. However, differences between barns in ammonia concentration, relative humidity, light and average barn temperature were very small, and showed no apparent relationship with flock stocking density. Although it is unlikely that differences in feather pecking between treatments can be entirely explained by the differences in light intensity measured outside the nest boxes, it is possible that, with increased stocking density, the light intensity on bird level was reduced, minimising the attractiveness of feathers and thereby reducing the level of feather pecking compared to the low stocking density. In contrast, differences in light intensity *inside* the nest boxes was an integral part of the modified management regime, and was clearly associated with reduced feather pecking in the high stocking density treatments.

The birds in the current study showed a higher incidence of feather pecking in the low compared to the high stocking density in the initial phase of the laying cycle (at 32 weeks of age). This is in contrast with the positive relationship between feather pecking and stocking density found in previous studies (Hansen and Braastad, 1994; Huber-Eicher and Audige, 1999; Nicol et al., 1999). Also, the effect of different stocking densities on feather pecking in our study could not be attributed to confounded changes in flock size (Appleby et al., 1989; Nicol et al., 1999) because our statistical comparison between the low and high stocking density was made at a similar small flock size. The unexpected conclusion that feather pecking and aggression were higher at low stocking densities is strengthened by an informal comparison of all six treatments (Fig. 2). Although confounding factors have to be considered in this comparison, feather pecking and aggression were generally higher in the low stocking density treatments than in treatments T3, T4 and T6, which were not included in detailed formal analysis of the effects of stocking density. A possible explanation is that relationships between stocking density and feather pecking alter as flock sizes increase. In previous studies the average flock size was about 350 birds (Appleby et al., 1989; Hansen and Braastad, 1994; Nicol et al., 1999) whereas our smallest flock size was 2450 birds. However, epidemiological surveys of comparable commercial flocks reported either no relationship between feather pecking and stocking density (Gunnarsson et al., 1999; Oden et al., 2002) or a positive one (Nicol et al., 2003; Huber-Eicher and Audige, 1999).

Another possibility is that the type of husbandry system could have accounted for the unusual relationship between stocking density and feather pecking. Systems in studies that found a positive relationship were floor pens (Hansen and Braastad, 1994; Huber-Eicher and Audige, 1999) and a perchery (Nicol et al., 1999), whereas in the current study birds were housed in a one-tier aviary system. Although Nicol et al. (2003), in a matched

case–control study of free-range farms, found that the occurrence of feather pecking was not associated with any particular husbandry system, differences in management practices within systems could be important, and may have been confounded with stocking density in previous epidemiological surveys.

In our experiment there was a marked lower incidence of feather pecking under modified management conditions. The factors incorporated in the modified management were nipple drinkers instead of bell drinkers and no lights on in the nest boxes. Our results confirm previous findings that these modifications decrease the risk of feather pecking (Green et al., 2000; Pöttsch et al., 2001). Whether and to what extent these management factors interact with effects of stocking density in general and on feather pecking in particular remains to be elucidated. We also found that modified management was associated with lower levels of aggression. How the modifications under modified management conditions affected the level of aggression is not clear as the incidence of feather pecking and aggression does not always co-vary (e.g. Nicol et al., 1999). One possibility is that the more evenly spaced nipple drinkers may have reduced competition and frustration, in comparison with the bell-drinkers where birds tend to be closer together.

Initially, the incidence of aggression was also higher in the low stocking density compared to the high stocking density treatments. This might be the result of birds adopting a strategy of social tolerance (Hughes et al., 1997; Pagel and Dawkins, 1997; D'Eath and Keeling, 2003; Estevez et al., 2002, 2003). This strategy is usually associated with larger flock sizes (>100 birds, Hughes et al., 1997). It could be that birds in our study perceived their flock as smaller in the low density treatments than in the high density treatments and therefore showed more aggression under low than under high stocking densities. In support, birds showed more aggression in small flocks (2450 birds) compared to large flocks (4200 birds). However, the small flocks in our study were markedly larger than the largest flocks used in the above-mentioned experiments (700: Hughes et al., 1997; 120: D'Eath and Keeling, 2003; Estevez et al., 2002, 2003) and it seems remarkable that we still found differences in aggression between flock sizes of 2450 and of 4200. This might be the result of the difference in usable space per bird between small and large flocks. At the same stocking density, birds in large flocks have more space available due to the stochastic movements of the flock causing dynamic variation in stocking density across the barn (Cooper and Albentossa, 2003). Birds in our small flocks might have been physically hampered in accessing resources (food, water, nest, litter) and therefore showed more frustration-induced aggression. This frustration might also explain why we also found more feather pecking in the small than in the large flocks. These findings merit further research into social dynamics of large flocks in relation to aggression and feather pecking.

With increasing age, the initial effects of a strategy of social tolerance in our flocks might have been overruled by the effect of stocking density. Both feather pecking and aggression showed an increase with age, but only in the high stocking density treatments. High stocking densities are characterised by lower levels of space available per bird and therefore physical constraints on movement. This probably results in frustration and this might explain the higher levels of aggression and feather pecking under high densities. This constraint on movement is probably aggravated by a small flock size, especially under standard management conditions; small, high density flocks under standard management showed a marked increase in aggression, and to a large extent also in feather pecking,

compared to the small flocks under modified management and compared to the large flocks under both type of management (see Fig. 6). There is also the possibility that under high stocking densities feather pecking is more likely to spread through social transmission (McAdie and Keeling, 2002) than under lower stocking densities. Perhaps, this social transmission effect overruled the possible, initial effects of social tolerance.

On average, allopreening was observed more in small than in large flocks and more under standard than under modified management. Preening was observed more in small than large flocks, but only under standard management conditions. The small flocks had lower absolute space available than the large flocks. If birds were avoiding particular parts of the environment (e.g. walls or corners) they might be forced closer together in smaller flock sizes. As birds seem to prefer to be close to other birds while preening (Keeling, 1994) this might have facilitated their preening and allopreening behaviour. Modified management reduced the difference in preening behaviour between small and large flocks; possibly because reductions in feather pecking and aggression reduced the need to re-adjust plumage. It is also possible that some preening may have been displacement preening, indicating higher levels of comfort in modified management flocks.

In conclusion, the current study is one of the few that investigated the effect of stocking density, flock size and management on laying hen behaviour on a commercial scale using a replicated design. Behavioural observations in this study did not show that the welfare of laying hens was compromised by housing them at 12 birds m^{-2} , in comparison with birds housed at 9 or 7 birds m^{-2} in single-level non-cage systems. In general, it seems that a low space allowance, induced by either a smaller flock size or a higher stocking density, results in higher levels of feather pecking and aggression. Initially, this effect seems to be modulated by a strategy of social tolerance. Furthermore, simple modifications in management significantly and profoundly improved bird welfare. Our results suggest that it might be worth investigating the effects on feather pecking and aggression of initially house laying hens at high stocking densities and decreasing this stocking density with age.

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